University of Sulaimani

College of Science

Physics Department

Undergraduate Program

Nuclear Physics Lecture Note

2026-2027

By

Dr. Adil M. Hussein M. Saeed

Assistant Prof. in Nuclear Physics

CHAPTER ONE

Basic Concepts in Nuclear Physics

1.1 Terminology:

${}^{A}_{Z}X$ denotes a nuclide, a specific nucleus with $Z(atomic\ number)$ is several protons, and $A(mass\ number)$ is several nucleons (neutrons or protons). The symbol of the nucleus is X which is either a single letter as in uranium U or two letters as in copper Cu or plutonium Pu. An example of a nuclide is ${}^{238}_{92}U$.

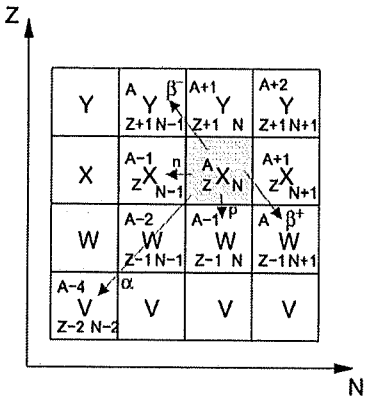
Symbol A is called the mass number, since knowing the number of nucleons, one has an approximate idea of the mass of the particular nucleus, such as ${}^{235}_{92}U$ and ${}^{238}_{92}U$.

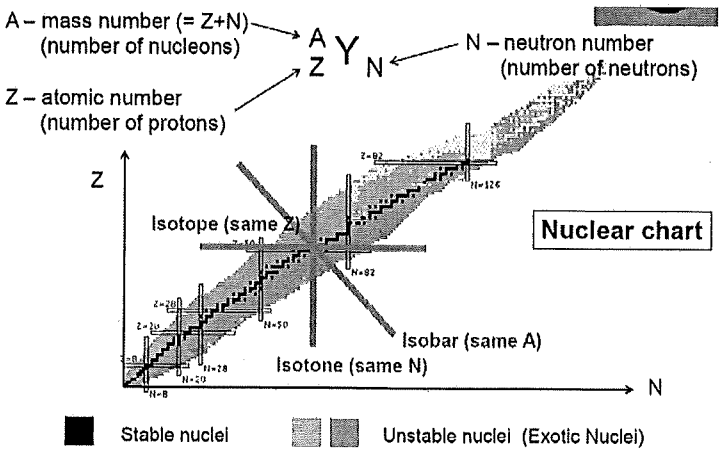
Isotopes are nuclides with the same Z but different A, e.g., ${}^{63}{}_{29}Cu_{34}$ and ${}^{65}{}_{29}Cu_{36}$.

Isotones are nuclides with the same N but different Z, e.g., ${}^{26}{}_{12}Mg_{14}$, ${}^{27}{}_{13}Al_{14}$, ${}^{28}{}_{14}Si_{14}$

Isobars are nuclides with the same A but different Z, e.g., ${}^{31}{}_{14}Si_{17}$, ${}^{31}{}_{15}P_{16}$, ${}^{31}{}_{16}S_{15}$.

Isomers: are nuclides with the same A and Z, but with extra excitation energy above the ground state, e. g., ${}^{99m}{}_{43}Tc$ ($t_{1/2}$=6h), ${}^{99}{}_{43}Tc$ ($t_{1/2}$=200,000y).





1.2 Nuclear Property

1.2. *a* - Nuclear Mass

The proton is approximately 1836 times as massive as the electron, and the masses of the proton and the neutron are nearly equal. ***The atomic mass unit*** (***u*** or **amu**) is defined so that the mass of one atom of the isotope ${}^{12}_{6}C$ is precisely **12** ***u***, so it regards as the unified (standard) scale. The proton and neutron each have a mass of approximately one ***u***.

Example: Use Avogadro's number to show that one atomic mass units (u) is equal to $1.660420x10^{-27}$ kg and express it in an energy unit.

Solution- part one: one atomic mass unit calculates as:

m =12 g =1 mol of ${}^{12}_{6}C$ contains Avogadro's number ($6.0223 * 10^{23}$) of atoms.

So, to find the mass ***m*** of one atom of ${}^{12}_{6}C$, which contains 12 nucleons of protons and neutrons.

$$m = 12 \times g/6.0223 * 10^{23} = 0.012 \frac{Kg}{6.0223*10^{23} atoms} = 1.99 * 10^{-26} Kg$$

amu (u) *unit convert to energy unit*:

$$1\text{amu of } C^{12} = \frac{m\left({}^{12}_{6}C\right)}{12} = \frac{1.99*10^{-26}Kg}{12} = 1.660420 \times 10^{-27}\text{kg}$$

, which is the mass of one nucleon of the nuclide ${}^{12}_{6}C$.

Part two: how is the mass of 1 u converted to the energy unit?

Solution: directly from $E = M\,c^2$

$$E_{(1\,u)} = 1.660420\times10^{-27}\text{kg}\left(3\times10^{8}\ \frac{\text{m}}{\text{sec}}\right)^2 = 1.494387\times10^{-10}\ \text{Joule}$$

$$E_{(1\,u)} = \frac{1.494387 \times 10^{-10}\ \text{Joule}}{1.602 \times 10^{-19}\ \frac{\text{Joule}}{\text{eV}}} = 931.48\ \text{MeV}$$

Typically, in nuclear science, mass is expressed in units of energy because nucleons exhibit continuous vibrational and rotational motions within a closed system known as a nuclide.

Q/ Why carbon-12 made standard constant for calibration mass atomic mass unit?

Ans.: Carbon-12 is most stable than Oxygen-16, and it has only one isotope. So, it is suitable for measuring the mass and its mass number (A) coincidentally with its atomic

mass unit (A = 12 = 12 amu). On another side, the mass (M) of one atom of any radionuclide is not equal to its atomic mass number (A), as apparent in this table.

	Z	A	Atomic mass (u)	I^π	*Abundance* or Half-life
H	1	1	1.007825	$\frac{1}{2}^+$	*99.985%*
		2	2.014102	1^+	*0.015%*
		3	3.016049	$\frac{1}{2}^+$	12.3 y (β^-)
He	2	3	3.016029	$\frac{1}{2}^+$	*1.38×10^{-4}%*
		4	4.002603	0^+	*99.99986%*
Li	3	6	6.015121	1^+	*7.5%*
		7	7.016003	$\frac{3}{2}^-$	*92.5%*
		8	8.022486	2^+	0.84 s (β^-)
Be	4	7	7.016928	$\frac{3}{2}^-$	53.3 d (ε)
		8	8.005305	0^+	0.07 fs (α)
		9	9.012182	$\frac{3}{2}^-$	*100 %*
		10	10.012937	3^+	*19.8%*
		11	11.009305	$\frac{3}{2}^-$	*80.2%*
		12	12.014353	1^+	20.4 ms (β^-)
		13	13.017780	$\frac{3}{2}^-$	17.4 ms (β^-)
C	6	9	9.031039	$\frac{3}{2}^-$	0.13 s (ε)
		10	10.016856	0^+	19.2 s (ε)
		11	11.011433	$\frac{3}{2}^-$	20.4 m (ε)
		12	12.000000	0^+	*98.89%*
		13	13.003355	$\frac{1}{2}^-$	*1.11%*
		14	14.003242	0^+	5730 y (β^-)
		15	15.010599	$\frac{1}{2}^+$	2.45 s (β^-)

Studies of atomic masses by the mass spectrograph shows that a nuclide has a mass nearly equal to the mass number A times the proton mass. Nuclides include three important particle proton, neutron and electron, which have rest mass values, in mass and energy units.

	mu [M(C^{12}) = 12]	Mev
electron	0.000548597	0.511006
proton	1.0072766	938.256
neutron	1.0086654	939.550

Q/ Why do we want to choose more stable nuclei?

Ans. In the bound system (nucleus), mass converts to energy and vice versa according to $E \rightleftarrows mC^2$. In stable nuclei, energy is not changed (low energy), cause to increase nucleus abundance ratio, so the change in mass is shallow.

Mass stability:

Nucleons are typically stable within the nucleus element (a bound system). The increase in mass is an indication of (inverse proportional with) the stability of the nuclide, this occurs when the motion of nucleons (protons or neutron) increased (excited), when the whole

nucleus affected by the external force (energy), such as bombarded the target by the accelerated particles (proton, neutron, electron, alpha and gamma-ray).

Neutron and Proton: The higher mass is equivalent to the higher the energy (less stable). Notice that the proton is lighter than the neutron, so the proton is more stable than the neutron. Indeed, if the neutron is not bound in a nucleus (that is, it is a free neutron), it will decay into a proton plus an electron (plus an antineutrino) with a half-life of about 13 min.

$Neutron \Longrightarrow Proton + Electron + anti_{Neutrino}$

Example:

$^{234}_{90}\text{Th} \rightarrow {}^{234}_{91}\text{Pa} + e^- + \text{anti-neutrino}$
thorium protactinium

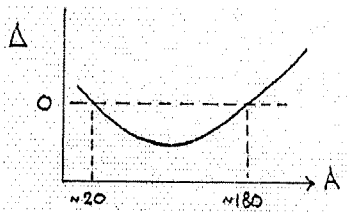
Q/ give the evidence for the neutron instability compared with the proton?

Note: Nuclear masses have been quite determined to high accuracy, the precision of 1 part in 10^8, by the methods of mass spectrography and energy measurements in nuclear reactions.

Increasing the mass defect increases energy, and the nucleus is less stable.

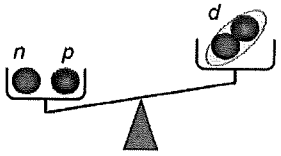
Mass defect is the difference between the actual mass (sum of an individual nucleons) of a nuclide and its mass number,

Δm = (total mass of individual nucleons) – (mass of nucleus)
$\Delta m = M - A$



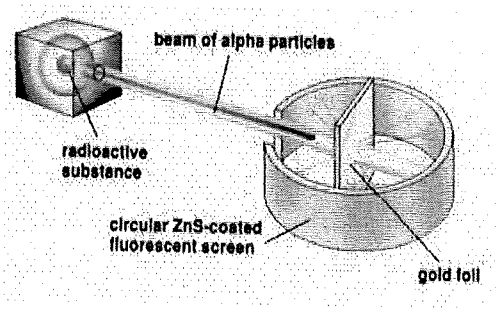
It is also called the "**mass decrement**". If we plot Δ *versus* A, we get a curve sketched in bellow Figure. When $\Delta < 0$, it means that the nucleus mass is lighter than the sum of the (individual nucleons) components. In other words, the nuclide is stable. Conversely, $\Delta > 0$ means the nuclide is unstable.

Our sketch shows that very light elements ($A < 20$) and heavy elements ($A > 180$) are not stable, which maximizes stability around $A = 56$.



Q / calculate the mass defect of alpha particles (He nucleus), return to page3 for getting (alpha, neutron and proton)

1.2. *b* Nuclear Size



In addition to J. J. Thomson's discovery of the atom in 1900, there are scattering experiments that help determine the distance to the nucleus. The scattering of alpha particles with gold foils (Geiger and Marsden, 1909) revealed a much greater degree of backscattering than would be expected from a simple diffusion process. Rutherford observed that this pattern suggested the influence of the alpha particle's electric (Coulomb) force. He subsequently formulated the law of scattering. Later measurements indicated that this law does not hold when

a- The alpha-particle kinetic energy is too high.

b- The atomic number of the scattered is too low.

When an alpha particle bombards the gold foil, the alpha particle is very distant from the nucleus of the gold, and it only has kinetic energy T_α. *It comes closest to the nucleus in a head-on collision.* At that point, the alpha particle has only potential energy if the recoil of the nucleus is neglected. Hence, by conservation of energy,

$$T_{\alpha(before\ collision)} + \underbrace{V_{\alpha(before\ collision)}}_{\cong 0} = \underbrace{T_{N(atfe\ rcollision)}}_{\cong 0} + \underbrace{V_{(after\ collision)}}_{\frac{2eZe}{D}}$$

$$T_\alpha = \frac{2eZe}{D} \qquad \text{(in electrostatic units)}$$

where $2e$ is the charge of the alpha particle ($e = 4.80 \times 10^{-10} esu$), Ze is the charge of the scattering nucleus, and D is the distance of closest approach.

$$D = \frac{2eZe}{T_\alpha}$$

For example, alpha particles show deviations from pure Coulomb scattering on uranium beyond $25MeV$). In that case,

$$D = \frac{2 \times 92 \times (4.8 \times 10^{-10} esu)^2}{25 \times 10^6\ eV} = \frac{184 \times 23.04 \times 10^{-20} esu^2}{25 \times 10^6 \times 1.602 \times 10^{-12}\ erg}$$

$$= \frac{4239.36 \times 10^{-20}\ erg\ cm}{25 \times 1.602 \times 10^{-6}\ erg} = 10.585 \times 10^{-13} cm = 10.585\ F$$

Where $esu^2 = \text{erg cm}$, $(1F_{(Fermi)} = 10^{-13} cm)$.

This experiment results; the nuclear radius to vary with the 1/3 power of the mass number, $R = r_0 A^{1/3}$, with $r_0 \sim (1.2 - 1.4) \times 10^{-13} cm$. The lower value (1.2 F) of the coefficient r_0 comes from electron scattering, which probes the charge distribution of the nucleus, while the higher value (1.4 F) comes from nuclear scattering, which probes the range of the nuclear force. The nuclear radii are conventional to adopt a length unit called $Fermi\ (F), 1F = 10^{-13} cm.$

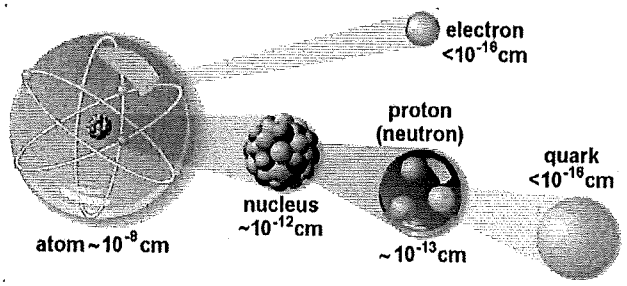
$$V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi\left(r_0 A^{1/3}\right)^3 = \frac{4}{3}\pi {r_0}^3 A \Rightarrow V \propto A$$

$$r_0 = \begin{cases} 1.4\ F \text{ for nuclear particle scattering (like fast neutrons)} \\ 1.2\ F \text{ for electromagnetic particle scattering (like electrons)} \end{cases}$$

$$\rho = \frac{\sum of\ nucleons N mass \in the\ nucleus}{Volume\ of\ the\ nucleus} = \frac{A * 1\ nucleon\ mass}{\frac{4}{3}\pi {r_0}^3 A} = \frac{3 * 1 nucleon\ mass}{4\pi {r_0}^3}$$

Then for the constant radius $r_0 = 1.4\ F$ we have:

$$\rho = \begin{cases} 0.087 \text{ nucleon } F^{-3}\text{: nucleon number density} \\ 1.45 \times 10^{17} kg.m^{-3}\text{: mass density, } where\ m_p = 1.67 \times 10^{-27} kg \end{cases}$$

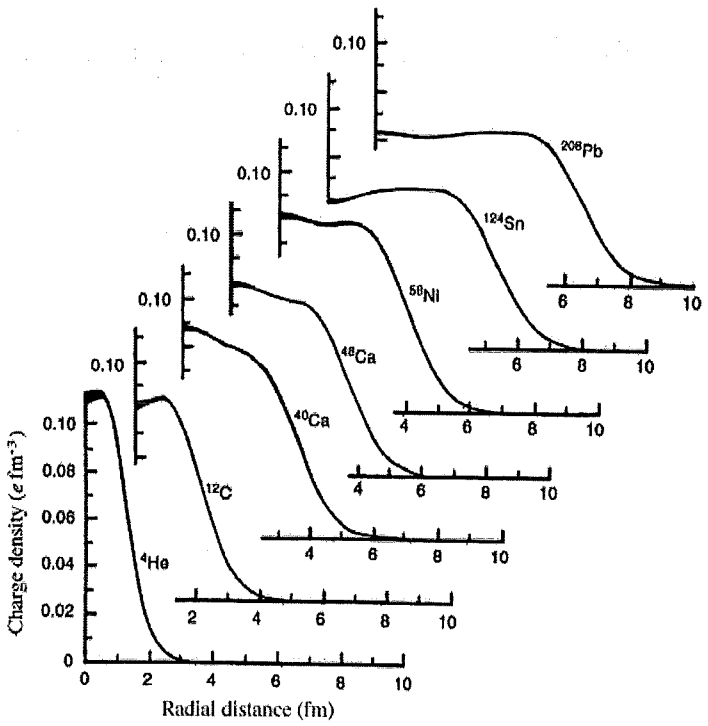


Q (H.W): Prove that the nuclear density is about $10^{38}\ nucleons/cm^3$.

1.2. c - Nuclear charge

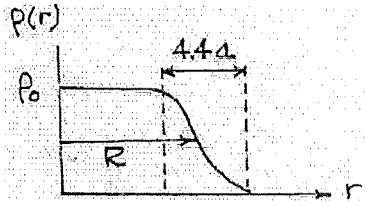
The charge of a nuclide ${}_zX^A$ is positive and equal to Ze, where(e) is the magnitude of the electron charge, e = **4.80298 x 10^{-10} esu (=1.602189 * 10^{-19} Coulomb)**. We consider single atoms as exactly neutral, the electron _ proton charge difference is $5 * 10^{-19} e$, and the charge of a neutron is $2 \times 10^{-15} e$.

From this Fig., two typical nucleon density distributions were obtained by high-energy E.S scattering. One can see two essential components in each distribution, **a core region** of constant density and a **boundary region** where the density decreases smoothly **to zero**.



The shape of the distributions shown in Fig. below can be fitted to the expression, called the Saxon distribution;

$$\rho(r) = \frac{\rho_o}{1+\exp[(r-R)/a]},$$

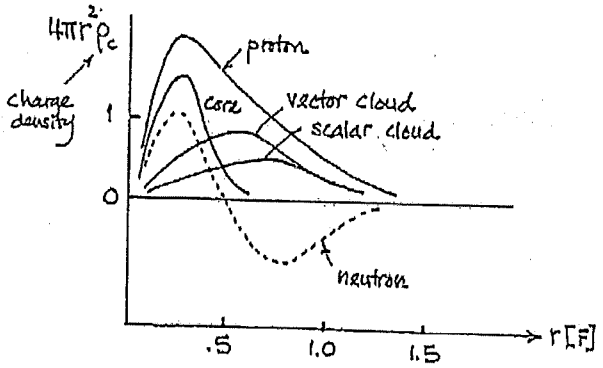


where $\rho_o = 1.65 \times \frac{10^{38} nucleons}{cm^3}$,

A sketch of this distribution, given in the figure below, clearly shows the distribution's core and boundary components. The magnitude of the nuclear density is 10^{38} nucleons per cm^3, compared to the atomic density of solids and liquids, which is in the range of 10^{24}nuclei per cm^3.

Detailed studies based on high-energy electron scattering have also revealed that even the proton and the neutron have complicated structures. This is illustrated in the figure below.

The mesons are unstable particles of mass between the electron and the proton: π_mesons (pions) play an essential role in nuclear forces $(m_\pi = 270m_e)$, μ mesons (muons) are essential in cosmic ray processes $(m_\mu = 207m_e)$.



Question: The radius of light nuclei is (A < 10) about **3 f_m**, but in heavy nuclei, (A > 200) is about **10 f_m**. Enhance your answer by equation.

Problem1:

A. An alpha particle of energy $T_\alpha = 4.2\ MeV$ makes a head-on collision with a nucleus of atomic number $Z = 92$ and mass number $A = 238$. Calculate the distance of the closest approach, taking into account the **recoil** of the nucleus.

B. An 0.2 MeV proton makes a head-on collision with an alpha particle at rest. What is the distance of the closest approach (in F)?

C. If an alpha particle makes a head-on collision with a proton at rest, what must be its kinetic energy so that the distance of the closest approach is identical to case (b)?

Problem2:

Suppose that the density of nucleons ρ in a nucleus varies with radial distance r from the centre of the nucleus, as shown in the figure. What fraction of the nucleons lies in the surface region in the nuclei Al^{27}, Te^{125} and Po^{216} if $\rho_0 = 0.17\ F^{-3}, c = R = 1.1\ A^{\frac{1}{3}}\ F,\ a = 3.0F$? (This problem can be solved without evaluating any complicated integrals.)

